

Spencer Wallace

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EDUCATION

University of Washington

PhD Astrophysics

Seattle, Washington

August 2023

University of Arizona

BS Computer Science, Astronomy and Physics

Tucson, Arizona

May 2014

EXPERIENCE

Postdoctoral Researcher

University of Washington, Astronomy Department (Remote)

September 2023 – Present

- Deployed and optimized large-scale C++/CUDA simulation code ([ChaNGa](#)) on NVIDIA Grace Hopper CPU-GPU nodes at TACC, improving runtime scalability on up to 500+ GPUs and 30,000+ cores
- Profiled and debugged GPU memory and performance bottlenecks with NVIDIA tools, reducing stalls and improving throughput by a factor of 5x
- Collaborated with HPC specialists across physics, CS, and biology to adapt algorithms to diverse workloads

Research Assistant

University of Washington, Astronomy Department

October 2019 – August 2023

- Designed and implemented a parallel tree-based collision detection module in C++/MPI, enabling the first planet formation simulations with realistic-sized bodies, which resulted in three first-author publications
- Co-authored [paratreet](#), a toolkit for testing/tuning distributed tree traversal algorithms, which resulted in a coauthored [IEEE publication](#)
- Built a PyTorch-based generative diffusion model to synthesize simulation results, saving 900k+ CPU hours
- Mentored 5 students in HPC workflows and performance tuning on clusters

Teaching Assistant

University of Washington, Astronomy Department

October 2015 – September 2019

- Led weekly discussion sections, graded exams, designed homework assignments and performed guest lectures for undergraduate astronomy classes, some of which involved teaching basic data analysis and numerical modeling skills using Python and Excel
- Collaborated with a team of other teaching assistants to ensure exams and assignments were graded consistently and fairly

Software Consultant

LSST Corporation

June 2011 – August 2014

- Designed interactive visualization widgets (JavaScript/Node.js) for [ASCOT](#) the astronomer's collaborative toolkit
- Built a fully customizable, persistent dashboard interface, along with documentation to facilitate large survey data exploration

SKILLS

Programming Languages: C++, Python, Bash, JavaScript

Tools and Environments: Git, CUDA, NumPy, pandas, PyTorch, SLURM, matplotlib, seaborn, node.js

Communication: 3 first-authored publications, 2 co-authored publications, 8 conference talks, 3 conference posters, 20 pop-sci articles published through [astrobites](#)